Ranking Under Biased Missingness: When Model Rank Is Not Identifiable from Sparse Leaderboards

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Abstract

A leaderboard rank implicitly claims that the reported scores order two models; on sparse leaderboards, that claim is usually taken for granted rather than checked. Public robotics and embodied-AI leaderboards report scores for only subsets of benchmarks. We study a cross-benchmark matrix spanning LIBERO, CALVIN, Meta-World, and LIBERO-Plus with 162 deduplicated models, 18 benchmark items, and 781 observed scores (26.8% density). Reporting coverage is negatively correlated with fitted item difficulty ($r = -0.639$), but this association does not identify the reporting mechanism. We fit a penalized two-parameter logistic item-response model and bootstrap observed cells ($B = 1000$). The resulting intervals quantify pipeline sensitivity, not coverage-guaranteed uncertainty: 36.4% of model pairs are unresolved by the pairwise criterion on the unmasked matrix. The maximum divergence between latent-trait and average-based ranks is $\Delta = 77$; the corresponding model's rank interval spans 58 positions. Under additional masking at 50% and 90%, we distinguish a fixed-denominator rate that includes unavailable pairs from a rate conditional on surviving models. At 90% masking these rates are 91.6% and 45.6%, respectively. Rank self-consistency with each estimator's own unmasked ranking also changes ordering: $\Delta\rho = -0.0031$ at 50% masking and $\Delta\rho = +0.106$ at 90%. These are outcomes for the sampled masks, not evidence of a stable regime boundary or superior ranking accuracy. Before a sparse ranking is trusted, it should report coverage and state which comparisons it can make, which it cannot resolve, and which it cannot make at all.

1 Introduction

A leaderboard rank is an implicit claim that the reported evidence orders two models. When every model is evaluated on every benchmark, that claim can be checked with standard uncertainty tools; when models report different subsets, whether the ordering can be trusted is usually taken for granted rather than checked.

Robotic foundation models, including vision-language-action (VLA) systems, are now compared across many partially overlapping manipulation benchmarks [Brohan et al., 2023]. LIBERO reports four language-conditioned manipulation suites [Liu et al., 2023], CALVIN reports average long-horizon task completions [Mees et al., 2022], Meta-World evaluates multi-task manipulation [Yu et al., 2020], and LIBERO-Plus measures robustness under perturbations [Fei et al., 2025]. Public leaderboards make this progress visible, but they also create a measurement problem: most models report only a subset of items, and easy suites saturate near 100% success.

The standard response to leaderboard noise is to attach uncertainty to reported scores and ranks [Neuhof and Benjamini, 2026b, Miller, 2024]. Hierarchical rank intervals distinguish within-task

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uncertainty from variation across tasks. The leaderboard-level construction of Neuhof and Benjamini [2026b] assumes that all models are evaluated on each task. Public robotics leaderboards instead contain absent model–benchmark scores, so uncertainty about the observed scores does not by itself address the incompleteness of the comparison set.

We take the sparse reported-score matrix without imputing missing scores, administering anchors, or restricting to a dense submatrix, and examine the sensitivity of a fitted latent-trait ordering. The title's identifiability language refers to an ordering not resolved by this available-data analysis; we do not prove statistical non-identifiability of the latent-trait model. Failure to separate a pair under the stated rule is not evidence that the models are equivalent.

Contributions.

1. We document an association between reporting coverage and fitted item difficulty ($r = -0.639$) on a public cross-benchmark VLA matrix, without identifying the selection mechanism.
2. We report rank and pairwise sensitivity intervals conditional on the observed cells, with 36.4% of pairs unresolved before additional masking.
3. We distinguish unresolved surviving pairs from unavailable pairs, reporting both the original-population and surviving-population denominators, and show that most models surviving severe masking rest on a single observed score.
4. We compare each estimator's self-consistency with its own unmasked ranking and report the observed sign change without claiming accuracy or a stable crossing point.

2 Related Work

Rank intervals for leaderboards. Neuhof and Benjamini [2026b] construct task-level rank confidence intervals from pairwise comparisons under family-wise error control and leaderboard-level rank prediction intervals through conformal aggregation. The latter address rank on a new task, not uncertainty about a single latent-trait ordering. Their leaderboard-level setup evaluates all models on each task. A companion analysis [Neuhof and Benjamini, 2026a] examines sources of ranking uncertainty on MMLU, including subject variation. Our cell-resampling analysis concerns a different source of sensitivity: which reported model–item cells support the fitted ordering. It neither inherits their coverage guarantees nor establishes that our interval procedure is preferable.

Partial evaluation matrices and anchor-based correction. Habba et al. [2026] use multidimensional IRT, anchor items, and fixed-parameter calibration to extend benchmark suites while maintaining comparability. That design supplies common measurements to link evaluations. We instead analyze an already-published matrix without administering a common anchor set. This is a difference in available evaluation design, not a demonstration that anchor-based correction is inferior or impossible. Shared observed items may still link models indirectly.

IRT for benchmark analysis. Classical IRT relates performance to latent ability and item parameters [Birnbaum, 1968, Lord, 1980]. Rodriguez et al. [2021] use IRT to study variation in the informativeness of NLP evaluation examples, and Lalor and Rodriguez [2022] provide scalable Bayesian estimation. Related work studies neural item models [Zhou et al., 2026], reduced evaluation sets [Polo et al., 2024], and structure-aware models for binary and continuous scores [Liao et al., 2025]. Prior work examines interactions between sparse evaluation, difficulty variation, and ranking error [Kang, 2026]. Our analysis uses suite-level aggregates and a unidimensional 2PL working model; this choice is not validated by the availability of aggregate scores alone.

Positioning. We audit a reported-score matrix through conditional ranking sensitivity, explicit comparison denominators, and exploratory self-consistency under additional masking, rather than propose a replacement ranking.

3 Method

3.1 Data

We load public JSON records from a VLA leaderboard aggregator [MINT-SJTU, 2026] covering four benchmark families: LIBERO, CALVIN, Meta-World, and LIBERO-Plus. Model names are normalized by lowercasing, stripping parenthesised variant suffixes, and preferring the leaderboard base name; on a repeated (model, item) pair the first observation is kept. This yields 162 deduplicated model identifiers and 18 benchmark items over 781 observed model–item scores, a density of 0.268. Cross-benchmark reporting is sparse: 141 models report at least two items, 124 report at least four, and 32 report at least seven — of which 31 span two or more benchmark families.

Scores are mapped to $[0, 1]$ per family: LIBERO, Meta-World, and LIBERO-Plus percentages are divided by 100 when the raw value exceeds 1, and CALVIN average task completion is divided by 5. All values are clipped to $[0.001, 0.999]$. Continuous suite-level scores are treated as pseudo-Binomial observations with $K = 100$ trials, which approximates reported success rates without pretending to recover instance-level rollouts. Missingness is encoded as `NaN`; missing reported-score cells are not imputed. The separate handling of absent bootstrap abilities is described in Section 3.3.

3.2 Latent-trait fit

For model i and item j, the 2PL item characteristic curve is

$$P(y_{ij} = 1 \mid \theta_i, a_j, b_j) = \sigma(a_j(\theta_i - b_j)) \,, \tag{1}$$

where θ_i is latent model ability, a_j is discrimination, b_j is difficulty, and σ is the logistic function. For an observed score s_{ij} in the set Ω of observed cells, the pseudo-Binomial data term is

$$\sum_{(i,j)\in\Omega} w_{ij}\Big[K s_{ij} \log p_{ij} + K(1 - s_{ij}) \log(1 - p_{ij})\Big], \tag{2}$$

with w_{ij} a per-cell weight (unity for the point fit; see Section 3.3). The estimator maximizes this term minus penalties applied in alternation: a ridge $0.5 \sum_i \theta_i^2$ together with a mean-centering term $10(\bar{\theta})^2$ on the ability step, and ridges $0.1 \sum_j b_j^2$ and $0.5 \sum_j (\log a_j)^2$ on the item step.

Estimation alternates penalized numerical optimization — one L-BFGS-B solve for θ, then one for $(b, \log a)$, each capped at 300 inner iterations — with analytic gradients at both steps. Ability is re-centered after every inner solve. Discrimination is bounded by clipping $\log a$ to $[-2.0, 1.6]$, giving $a \in [0.135, 4.953]$. Initialization is cold at every call: $\theta \sim \mathcal{N}(0, 0.3^2)$, $\log a \sim \mathcal{N}(0, 0.2^2)$, and b_j at the negative logit of item j's clipped observed mean. The outer loop terminates when the change in the ability objective falls below 10^{-5}, capped at 2000 iterations for the point fit and 1000 for each bootstrap replicate; replicates that reach the cap (0.6%, 0.2%, and 1.4% at 0%, 50%, and 90% masking, respectively) are kept. The canonical fit converged in 197 outer iterations to a final change of 9.2×10^{-6}. The shared latent axis is a working assumption; numerical convergence establishes neither model fit nor validity under selective reporting.

3.3 Rank intervals and indistinguishability

These percentile ranges are *pipeline sensitivity intervals*, not coverage-guaranteed confidence intervals. Resampling observed cells neither accounts for all shared model and item effects nor models the reporting mechanism. Pairwise classifications carry no simultaneous error guarantee and do not establish model equivalence or superiority.

We bootstrap over *observed cells*, not over models or items: each of $B = 1000$ replicates draws n indices with replacement from the n observed model–item cells and converts them to per-cell multiplicities w_{ij}, refitting the full 2PL under those weights. A model absent from a replicate (no cell with positive weight) receives $\theta = \texttt{NaN}$ for that replicate.

Ability intervals are naive 2.5/97.5 percentiles over the non-`NaN` replicates. Rank intervals are percentiles of the replicate-wise re-ranking, rounded outward ($\lfloor\cdot\rfloor$ lower, $\lceil\cdot\rceil$ upper); rank 1 is highest, and interval width is the upper minus the lower endpoint. For a pair (u, v) we take the percentile interval of $\theta_u - \theta_v$ over replicates containing both; the pair is called *distinguishable* if that interval

Table 1: Selected item parameters. The upper block contains the highest-discrimination items; the lower block provides comparisons. Coverage is the fraction of the 162 models reporting the item.

Item	a	b	Coverage
Meta-World-Very-Hard	4.953	-0.075	0.043
Meta-World-Hard	4.354	-0.083	0.068
LIBERO-Plus-Light	4.090	-0.343	0.154
Meta-World-Medium	3.600	-0.055	0.068
LIBERO-Plus-Background	3.588	-0.410	0.154
LIBERO-Plus-Noise	3.540	-0.068	0.154
LIBERO-Long	2.596	-0.773	0.784
CALVIN-D	0.894	-1.175	0.080

excludes zero and *indistinguishable* otherwise; below we also call indistinguishable pairs *unresolved*. A pair involving a model dropped by masking has no computed interval and is *unavailable*; in the reported runs every surviving pair has one. All randomness is seeded: a master seed of 20260801 is combined with a CRC32 of the condition label to give a per-condition seed, and the fit initialization seed is fixed at 0 for the point fit and every replicate, so replicates differ only through w.

For rank intervals only, absent models receive their point-estimate abilities before re-ranking. This suppresses their ability variation, although their ranks can still move as other models change. Pairwise difference intervals instead use only replicates containing both models, without these substitutions. Neither construction establishes that the reported indistinguishability rates are conservative.

3.4 Masking sweep

Masking is applied *on top of* the natural missingness, never to a denser submatrix. The candidate pool for hiding is exactly the set of already-observed cells; cells that are missing in the public data are never candidates and are never filled. At mask rate m we hide $\lfloor m \cdot n \rfloor$ observed cells uniformly without replacement, then drop any row or column left with no observations. We sweep $m \in \{0.0, 0.5, 0.9\}$, hiding 0, 390, and 702 cells and leaving 781, 391, and 79 observations respectively. The full bootstrap is rerun inside each condition. Only a single random mask is used at each level, so the sweep describes the sampled realizations rather than an average over masks. Random masking of observed cells does not reproduce or identify the original reporting mechanism.

4 Experiments

4.1 Reporting coverage is associated with fitted difficulty

Across the 18 items, reporting coverage and fitted difficulty are negatively correlated, $r = -0.639$. The structure is stark by family: the four LIBERO suites are reported by 75–78% of models, the seven LIBERO-Plus perturbation items by 15.4% each, and the four Meta-World difficulty strata by 6.8%, falling to 4.3% for Meta-World-Very-Hard. Table 1 includes the highest-discrimination items and comparison items.

Exactly one item sits at the discrimination bound (Meta-World-Very-Hard, $a = 4.953$), and it is the least reported item in the matrix. The five next-most discriminative items are all Meta-World or LIBERO-Plus items at 6.8–15.4% coverage, while LIBERO-Long, the hardest LIBERO suite by fitted difficulty, discriminates at 2.596 with 78.4% coverage. The highest-discrimination items are also among the least reported; this association does not identify why those results are absent. Difficulty is fitted from the same selected scores, and benchmark-family differences may contribute to the correlation. Sparse coverage and the discrimination bound also limit interpretations of item informativeness.

4.2 Pairwise sensitivity before additional masking

On the unmasked matrix, 4743 of 13041 model pairs (36.4%) are unresolved by the pairwise sensitivity criterion. This is a property of the fitted pipeline and resampling rule, not a coverage-

Table 2: Masking sweep for the sampled masks. Of the $N = 13041$ original pairs, E survive; survivors are separated (D, the pipeline's "distinguishable") or unresolved (R), and the remaining A pairs are unavailable because a model has no remaining observation, so $D + R + A = N$. Survival denotes numerical inclusion, not verified identification.

	mask 0%	mask 50%	mask 90%
Observations remaining	781	391	79
Models fitted	162	147	64
Items fitted	18	18	16
Surviving pairs, E	13041	10731	2016
Separated pairs, D	8298	6009	1097
Unresolved surviving, $R = E - D$	4743	4722	919
Unavailable, $A = N - E$	0	2310	11025
Unavailable or unresolved, $(A + R)/N$	36.4%	53.9%	91.6%
Unresolved among survivors, R/E	36.4%	44.0%	45.6%

Table 3: Self-consistency with each estimator's own unmasked ranking, not accuracy, for the sampled masks. $\Delta\rho = \rho_{\text{IRT}} - \rho_{\text{mean}}$, computed from unrounded correlations.

	mask 0%	mask 50%	mask 90%
ρ_{IRT}	1.000	0.922	0.820
ρ_{mean}	1.000	0.925	0.713
$\Delta\rho$	0.000	$\mathbf{-0.0031}$	$+0.106$

guaranteed fraction of equivalent models. Every one of the 4743 has a nonzero point difference in ability; none is a tie. The largest such case has $|\Delta\theta| = 1.070$ with a central 95% sensitivity interval of $[-0.059, 1.767]$ for the signed difference.

Rank intervals are correspondingly wide. Widths at zero masking range from 2 positions (the top-ranked model, interval $[1, 3]$) to 148 (a model ranked 137th with interval $[10, 158]$). Among the 78 models reporting four items (the median number of items per model), the median rank-interval width is 34.5 positions.

The largest divergence between latent-trait rank and average-based rank is $\Delta = 77$, and the corresponding model (IRT rank 22) has a rank sensitivity interval of $[3, 61]$, spanning 58 positions.[1] The divergence measures disagreement between estimators, whereas the interval describes variability of the latent-trait rank; neither establishes which ordering is correct.

4.3 Unavailable versus unresolved comparisons

Table 2 separates unavailable comparisons from unresolved surviving pairs. Let N denote original pairs, E surviving pairs, and D pairs separated by the sensitivity criterion. The fixed-denominator rate is $(N - D)/N$; the estimable-pair rate is $(E - D)/E$. The former includes unavailable pairs by convention; the latter conditions on survival.

At 90% masking, these rates are 91.6% and 45.6%. Their difference results from including dropped-model pairs and changing the denominator; it is not itself a count or proportion of unavailable pairs. In counts, 11944 pairs are unavailable or unresolved under the fixed denominator: 11025 involve a model with no remaining observations, and 919 are surviving pairs that fail the separation rule. The unavailable share alone is 84.5% of original pairs. Across the sweep, the estimable-pair rates are 36.4%, 44.0%, and 45.6%, but the models contributing to those denominators also change.

The estimable-pair rate is a conditional sensitivity diagnostic; the fixed-denominator rate also includes comparison unavailability. An unavailable pair has no fitted comparison. An unresolved surviving

[1] Code, data snapshots and verification scripts: `https://github.com/testofschool/vla-mask-verdict` (commit `30f7130`). The reported unmasked unresolved-pair rate is 36.4% on macOS arm64 and 36.3% in an independent Linux x86_64 reproduction.

pair has a computed comparison that fails the separation rule. Neither category establishes model equivalence.

A leaderboard can look simpler after poorly covered models disappear without providing stronger evidence. Reporting both rates and the surviving population makes this selection visible; changes in the conditional rate are not a within-pair masking effect.

Survival itself can rest on very little evidence. At 90% masking, 52 of the 64 surviving models have a single observed score. Under the cell bootstrap, a model observed in one of $n = 79$ cells is absent from a replicate with probability $(1 - 1/79)^{79} \approx 36.6\%$; across the 52 single-score models, the realized absence rate averages 36.7%. A model that remains on a sparse leaderboard is therefore not necessarily supported by a broad set of comparisons.

4.4 Self-consistency depends on the sampled mask

Self-consistency under masking does not favour the latent-trait estimator at every level. Each estimator is scored against its own unmasked ranking, so ρ_{IRT} and ρ_{mean} measure how well each recovers what it itself produced on the full matrix. These values are therefore stability diagnostics, not accuracy against a known ground-truth ranking.

For the sampled mask at 50%, simple averaging is marginally more self-consistent with its own unmasked ranking ($\Delta\rho = -0.0031$). For the sampled mask at 90%, the ordering reverses ($\Delta\rho = +0.106$). These values describe the realized masks; they do not establish a population-level ordering of the estimators.

Repeated masks are needed to assess whether this sign change is reproducible. The sweep does not locate a stable crossing point or establish accuracy: self-consistency can reflect stability around an estimator's own biased ranking.

5 Discussion

A simple average summarizes reported normalized scores; a latent-trait score summarizes them through an assumed ability scale and fitted item parameters. Neither automatically corrects selective reporting. The unresolved-pair fraction reported here applies to the latent-trait sensitivity analysis; we do not estimate an analogous uncertainty fraction for averaging.

The practical contribution is a reporting rule, not a replacement leaderboard. Show benchmark coverage beside each model's rank and rank sensitivity interval. For pairwise conclusions, use the interval of the paired ability difference rather than overlap of separate rank intervals. Distinguish comparisons separated by that diagnostic from unresolved surviving comparisons and unavailable comparisons, and state the denominator and interval construction. Non-separation is not equivalence.

Future leaderboards should preserve model-variant identities and evaluation protocols, publish per-task trial information, and provide shared measurements linking benchmark families. For this analysis, model-fit and dimensionality checks, repeated masks, and sensitivity to pseudo-trial weighting and reporting assumptions remain necessary validation steps. A shared anchor design is a future intervention, not a demonstrated result.

6 Limitations

This study uses public aggregates, not raw rollouts. Suite scores hide instance difficulty, seed variance, policy stochasticity, and protocol differences. The pseudo-Binomial likelihood with $K = 100$ is a working approximation, not recovered trial information. No goodness-of-fit, held-out prediction, or dimensionality check is reported. The shared latent scale and equal pseudo-trial weighting across heterogeneous aggregates remain unvalidated. The pseudo-trial count can interact with regularization, and no sensitivity sweep over K is reported.

Name-based matching may merge or separate model variants imperfectly. The coverage–difficulty association does not establish a missing-not-at-random mechanism: benchmark family, availability, and evaluation choices may contribute. Selection is not modeled, so θ may reflect both capability and reporting patterns; performance on unreported benchmarks is not identified by this analysis.

Cell-wise resampling treats observed cells as exchangeable despite shared effects. Rank intervals substitute point-estimate abilities for absent models; pairwise intervals condition on both models being present. Neither treatment proves conservative coverage or conservative indistinguishability rates. These are pipeline sensitivity diagnostics without simultaneous guarantees for pairwise decisions. No analysis excluding replicates that reached the iteration cap is reported.

Surviving rows define numerical inclusion, not verified identification. No diagnostic of model–item observation-graph connectivity is reported; regularization can produce an ordering without establishing support from linked observations. Sparse or weakly linked evaluations warrant caution. Masking changes the surviving population and uses a single realization at each level, so the sign change is exploratory rather than a stable crossing-point estimate.

7 Conclusion

A sparse leaderboard ranking can be trusted only as far as the comparisons behind it. In this matrix, 36.4% of model pairs are unresolved by an observed-cell sensitivity criterion before additional masking. Under severe masking, most original comparisons become unavailable rather than unresolved, and most surviving models rest on a single score. The sampled masks also yield different orderings of the estimators' self-consistency, without establishing accuracy or a stable regime boundary. Before a sparse ranking is trusted, it should publish coverage, the interval procedure, and both comparison denominators alongside point ranks.

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